Known but Unsayable: Decoding Defaults, Not Forgetting, Hide What Language Models Know

Anonymous ACL submission

Abstract

Language models are said to lose the long tail of knowledge. We separate knowledge a model lacks, holds but rarely says, and holds but cannot say because the sampler gives it probability exactly zero. Top-p, top-k and min-p sampling delete low-ranked tokens, so one deleted token makes a name impossible. We pair an exact per-token survival test from prior work with a recognition test against near-miss foils, run each deployed sampler exactly, with a numeric margin; 831,000 real samples never contradict it. Of the long-tail entities Qwen2.5-7B-Instruct recognises, 95.1% can never be produced under top-p 0.9 and 98.9% under its shipped sampler, against 52.4% for its base model. Post-training buries the tail and truncation seals it. Across OLMo-2's post-training stages the share rises from 54.4% to 93.8%, mostly at supervised fine-tuning, and the cut moves to the first token. Without truncation, the median deleted name would still be 151 times less likely than the base model's. Truncation adds a hard ceiling: across 277 categories, no amount of sampling finds more than 1,252 of 15,514 recognised names, so sample-based knowledge estimates measure that ceiling. Rarer entities are deleted more often. Relaxing truncation at the first token recovers only 12.3%; we report the frontier against invented answers.

1 Introduction

Ask Qwen2.5-7B-Instruct (Qwen et al., 2024) for a capital city of a U.S. state 1,000 times under the sampler it ships with, and it gives 4 different answers: Atlanta (557 times), Phoenix (235), Albany (127) and Denver (81). It knows many more. Shown a state capital beside three large U.S. cities that are not capitals, it picks the capital for 93.9% of the 49 capitals. It cannot say them. At the first step of the answer, nucleus sampling with $p = 0.9$ keeps only 7 candidate first tokens; the base model keeps 110. Honolulu starts with a token ranked 16th, behind tokens that together hold 96.4% of the probability, so it can never begin. Montpelier shares its first token with Montgomery. After that token the model puts 98.6% of the probability on Montgomery's next token, so even $p = 0.95$ never completes Montpelier. Under top-p 0.9 the base model can reach 49 of the 49 capitals and the chat model 8 (Figure 1).

The mechanism is general. Nucleus (top-p) sampling (Holtzman et al., 2020), top-k sampling (Fan et al., 2018) and min-p (Nguyen et al., 2025) keep only the most likely tokens at each step and delete the rest before sampling. If one token of a name falls outside the kept set at any step, the probability of the whole name is exactly zero. No number of samples, retries or self-consistency votes can produce it. Whether this happens can be tested exactly, with one teacher-forced pass per answer; Awad et al. (2026) introduced this per-token survival test for ordinary words in book continuations.

This matters for the debate on homogenisation and knowledge collapse (Peterson, 2025; Wright et al., 2025). When a model never produces a rare fact, three different things can be true: the model does not store it; it stores it but gives it a small probability, so enough samples will find it; or it stores it and the sampler deletes it. In practice the second and third cases combine. Post-training pushes most recognised names to tiny probabilities, and truncation then sets them to exactly zero: post-training buries the tail, and truncation seals it. The burial is a property of the model; the seal is a property of a decoding setting. From samples, both look like ignorance. The seal also has a cost downstream. Text written under these samplers can never contain such names, and training on generated text is known to erode the tails of a distribution (Shumailov et al., 2024; Guo et al., 2024).

We measure the seal exactly and the burial beneath it. Our contributions:

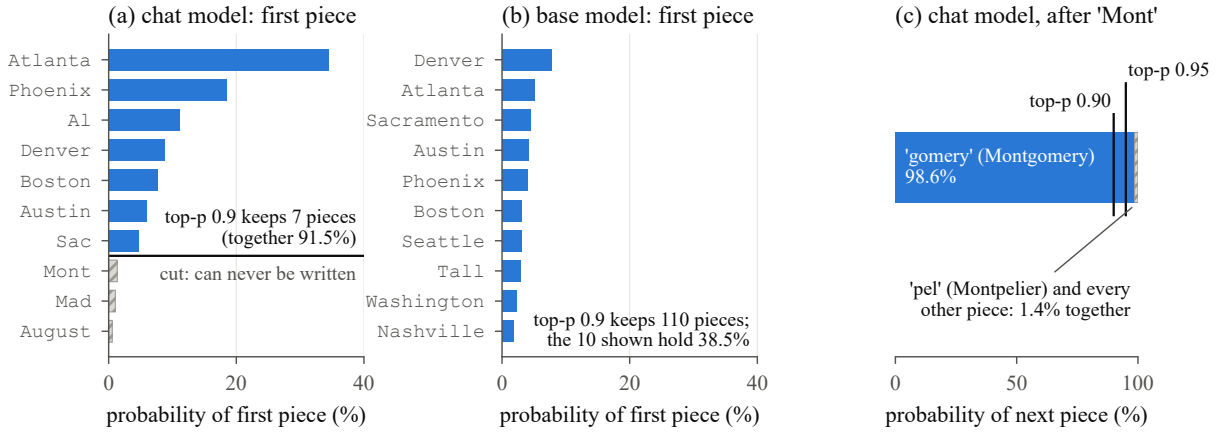


Figure 1: How a name becomes unreachable. Prompt: "Name a capital city of a U.S. state." (a) The chat model's ten most likely first pieces. Top-p 0.9 keeps the 7 pieces above the line (91.5% of the probability) and deletes the rest, so no capital starting with a deleted piece can be written. (b) The base model spreads its first step: its top piece has 7.8% and top-p 0.9 keeps 110 pieces. (c) A name can also be cut later. After the shared piece 'Mont', the chat model puts 98.6% on 'gomery' (Montgomery), so 'pel' (Montpelier) is deleted even at top-p 0.95. Hatched: deleted pieces.

1. **Recognised but unreachable.** We pair the per-token survival test of Awad et al. (2026) with a recognition test against near-miss foils. We run each model's deployed processor chain exactly and add a calibrated numeric margin. Real samples never contradict a robust verdict: 0 violations in 831,000 samples at full scale (Section 2).

2. **How much, at deployed settings.** Among entities Qwen2.5-7B-Instruct recognises, 98.9% of the pilot entities are unreachable under its shipped sampler, against 52.4% for its base model at top-p 0.9. The result holds on the categories where the recognition test is strongest and under the stricter all-wordings count (Section 4).

3. **Where it comes from.** Post-training is known to concentrate answers (Jang et al., 2026; Parikh, 2026). We link that concentration to exact zeros, stage by stage: from OLMo-2's base model through three post-training stages, most of the rise happens at supervised fine-tuning, and the cut moves to the first token of the name (Section 5).

4. **What it does to measurement.** Truncation caps what any amount of sampling can find at the reachable set. Sample-based knowledge estimators therefore see a ceiling, and below it a distribution that buries most recognised names. We split the gap to the recognised set into these two parts, using log-probabilities whose predictions match real samples (Section 6).

5. **Who loses, and what a fix costs.** Rarer recognised entities are deleted more often. Relaxing truncation at the first token recovers little, because names fail again later; we report the frontier between reachable knowledge and invented answers, with practical as well as formal recovery (Section 8). **[pdSelfCheckFix]**

2 Exact reachability

Samplers. At each decoding step a sampler applies, in the order used by Hugging Face `transformers` 5.22.2: repetition penalty, temperature T, top-k, top-p and min-p. Each step sets some logits to $-\infty$; sampling draws from the softmax of the rest. A token is *kept* if its processed logit is finite. Other inference engines order these steps differently (Limitations).

Keep rules. For a target token t with logit z_t: top-k keeps t iff fewer than k tokens have a larger logit; top-p keeps t iff $G(t) < p$, where $G(t)$ is the probability, at temperature T, of the tokens strictly more likely than t; min-p keeps t iff its probability is at least *min-p* times the top token's. Unit tests on random logits check that these rules agree with the library's processors.

Reachability of an answer. Write an entity e after prompt x as tokens $t_1 \dots t_L$ (the tokenisation

the model itself produces in context). e is reachable under sampler S iff every t_j is kept given x and $t_{<j}$. This is the per-token survival test of Awad et al. (2026). One teacher-forced pass gives every t_j's logits, so reachability is exact and needs no sampling. It also yields critical values: $p^*(e) = \max_j G_j(e)$ (e is reachable under top-p iff $p > p^*$), $k^*(e)$ and the critical min-p. For combined samplers (the shipped settings) we run the library's own processor chain at every position. An entity is reachable if any of its surface forms (label or up to three aliases) is.

A numeric margin. The same logits computed two ways (a batched teacher-forced pass, and key-value-cached generation) differ slightly in bfloat16, and bfloat16 logits often tie. A token on a truncation boundary can therefore be kept by one computation and cut by the other. We call e *robustly unreachable* if some token stays cut even after its logit is raised by ϵ relative to every other token. Every processor above is monotone in this shift, and the shift breaks all ties. ϵ is calibrated per model before each experiment. We sample 500 answers per pilot category under each of two samplers (11,000 answers, 1,685 distinct), compute for each the logit shift at which its verdict would flip, and take the smallest value on a fixed grid that covers every sampled answer (Appendix A). For Qwen2.5-7B-Instruct, $\epsilon = 0.75$; the largest relative logit gap among the tokens that decide a verdict was 0.94 logits. Headline numbers use the robust verdict, a lower bound; exact and upper-bound values are reported beside them (Appendix A).

Validation. We draw 1,000 real samples per category from `generate()` under each claimed sampler and check that no answer we call robustly unreachable is ever produced. On the pilot there were 0 violations in 22,000 samples for Qwen2.5-7B-Instruct and 0 in 33,000 for the OLMo-2 post-trained stages. On the full TailCat there were 0 in 554,000 samples for Qwen2.5-7B-Instruct and 0 in 277,000 for OLMo-2-Instruct. For Qwen2.5-0.5B-Instruct on CPU in float32, the two computations agree exactly and every sampled answer is scored reachable; we did not repeat this check for the 7B models.

3 TailCat and recognition

Categories. TailCat pairs categories of long-tail entities with near-miss foils from sibling categories, built from Wikidata (Vrandečić and Krötzsch, 2014): for example poets who wrote in Tamil (foils: Telugu and Kannada poets) or rivers in Kerala (foils: rivers in Karnataka and Tamil Nadu). Members are stratified over popularity (Wikipedia sitelinks) and must have articles in at least two languages, which removes single-wiki bot imports whose transliterated labels no reader could know. The pilot has 11 categories (2,312 entities) including a head control, US state capitals. The full TailCat crosses 14 entity types (people, places, works and organisations) with 24 countries, 17 of them in South Asia, Africa, Southeast Asia and Latin America, and adds writers in 12 Indian languages. Foils come from the same type in neighbouring or easily confused countries. We keep a category only if it yields at least 30 members and 30 foils: 277 categories and 33,437 sampled members (at most 150 per category).

Recognition. An entity counts as recognised if the model (i) answers "Yes" to 'Is "e" *category*?' with a score $V(e) = P(\text{Yes})/(P(\text{Yes}) + P(\text{No}))$ at or above the 95th percentile of the category's foils, so at most 5% of foils pass, and (ii) picks e out of three foils in a four-way choice. RBU is the share of recognised entities that are unreachable. The yes/no probe separates members from foils with AUROC 0.835 for Qwen2.5-7B-Instruct on the pilot and 0.847 on the full TailCat, below the 0.85 we set in advance (Appendix G). Two checks follow. On the 5 pilot categories where the probe reaches 0.85 on its own, RBU at top-p 0.9 is 98.2% for Qwen2.5-7B-Instruct and 54.5% for its base model; on the 139 such categories of the full TailCat it is 93.1% and 58.2%. With a second probe wording that adds worked examples, 1,234 pilot entities are recognised and RBU is 94.8% and 52.5% (Table 3). A third test asks the model for each name directly, with a question only that name answers: **[pdAskDirectly]**.

4 How much recognised knowledge is unreachable?

Table 1 reports RBU for the two families. Under the sampler Qwen2.5-7B-Instruct ships with (temperature 0.7, top-k 20, top-p 0.8, repetition penalty 1.05) 98.9% of the recognised pilot entities are unreachable (95% cluster-bootstrap CI 97.4% to 99.6%). Under plain top-p 0.9 it is 95.1% (87.8% to 98.1%), against 52.4% for the base model. On the full TailCat the numbers are 91.9% (90.4% to

	pilot		full TailCat	
	mean	all	mean	all
Qwen instruct, shipped	98.9%	98.6%	98.8%	98.5%
Qwen instruct, p 0.9	95.1%	94.0%	91.9%	89.7%
Qwen base, p 0.9	52.4%	47.0%	55.0%	50.1%
OLMo instruct, p 0.9	93.8%	91.9%	91.2%	88.5%
OLMo base, p 0.9	54.4%	47.0%	62.7%	56.3%

Table 1: RBU: share of recognised entities that are robustly unreachable. Qwen: Qwen2.5-7B; OLMo: OLMo-2-1124-7B; p 0.9: top-p 0.9 at $T = 1$. Mean: averaged over three prompt wordings (RBU-mean). All: unreachable under every wording (RBU-all). OLMo-2 ships no sampler (greedy decoding).

93.4%) for Qwen2.5-7B-Instruct and 91.2% for OLMo-2-Instruct, against 55.0% and 62.7% for their base models. Counting an entity only if it is unreachable under every wording (RBU-all, Table 1) gives lower values for every model and leaves every comparison intact. Figure 4 (Appendix D) shows how RBU falls as each truncation is loosened; it uses the exact verdict, which is slightly higher than the robust headline.

The effect is not a quirk of wording or temperature. In a separate run with five prompt wordings, the coefficient of variation of RBU at top-p 0.9 is 0.019 for Qwen2.5-7B-Instruct and 0.009 for OLMo-2-Instruct. Lowering the temperature raises RBU: for the Qwen base model from 52.8% at $T = 1$ to 85.2% at $T = 0.7$ and 92.0% at $T = 0.5$. In the same run, counting only the canonical label, not aliases, gives 94.8% against 94.1% with aliases.

Not a quirk of the one-name format. Users often ask for lists. **[pdListPrompt]**

Scale does not remove it. Across Qwen2.5-Instruct at 0.5B, 1.5B, 3B and 7B parameters the number of recognised pilot entities grows (328, 868, 878 and 1,310), and the share unreachable under each model's shipped sampler stays high (79.4%, 98.3%, 96.4% and 98.9%). The smallest model's recognition test is weak (AUROC 0.64), so its value is the least reliable.

Even head knowledge collapses. US state capitals were our pre-registered head control, which every model should reach. We judged the control on the base model (Appendix G), which reaches 49 of the 49 capitals in our pilot list under top-p 0.9. The instruct model reaches 8: the opening example.

5 Where it comes from

Post-training, stage by stage. OLMo-2 (Team OLMo et al., 2024) releases its base model and each post-training stage. RBU at top-p 0.9 is 54.4% for the base model, 89.6% after supervised fine-tuning, 93.8% after preference tuning and 93.8% for the final model. Most of the rise happens at supervised fine-tuning. Among entities recognised by both, 348 are unreachable only for the final model and 3 only for the base model (sign test, $z = 18.4$, $p < 10^{-74}$); for Qwen2.5 the counts are 456 and 5 ($z = 21.0$, $p < 10^{-97}$). Neither fine-tuning objective can set a probability to zero by itself: the KL-regularised preference optimum keeps the support of the reference model (Rafailov et al., 2023), and cross-entropy training keeps logits finite. What they change is where the mass goes (Figure 2). They bury the tail: on the full TailCat, the median recognised name that top-p 0.9 deletes from Qwen2.5-7B-Instruct would have probability 1.4×10^{-11} without truncation, against 2.1×10^{-9} for the median name the base model deletes (2.9×10^{-10} and 4.7×10^{-9} for OLMo-2).

The first token. Post-training is known to concentrate the answer slot: instruction-tuned models give one answer where a distribution is wanted (Jang et al., 2026), category prompts such as "Name a tree" converge across many models (Parikh, 2026), and output diversity collapses at identifiable post-training stages (Karouzos et al., 2026). Our test shows what this does to recognised knowledge. The share of unreachable recognised entities whose first failing token is the first token of the name rises from 32.4% (OLMo-2 base) to 73.8% (OLMo-2-Instruct), and from 28.0% to 84.1% for Qwen2.5. The mean probability of the single most likely first answer token rises with it, from 0.25 to 0.53 (OLMo-2) and from 0.19 to 0.52 (Qwen2.5). Across 66 (model, category) points, first-token concentration and RBU have Spearman correlation 0.82. This pooled value mixes the gap between base and chat models with the association inside each model. A post hoc check separates the two. Within each of the four models, over the 277 full-TailCat categories, the Spearman correlation is between 0.34 and 0.59, with every 95% CI above zero. Within a category, across OLMo-2's stages, the Pearson correlation is 0.82 (0.71 to 0.93). The association is real but weaker than the pooled value suggests. Appendix C shows real names token by token: the instruct model rules them out at the first

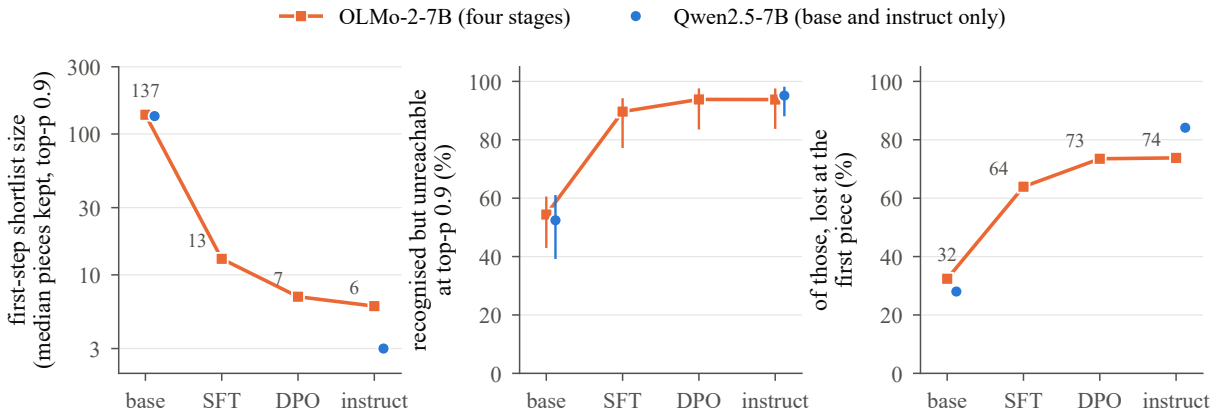


Figure 2: Where unreachability comes from: OLMo-2-7B after each post-training stage (pilot, 11 categories; SFT: supervised fine-tuning; DPO: preference tuning), with the Qwen2.5-7B base and instruct pair for comparison. SFT alone shrinks the first-step shortlist from 137 to 13 pieces. From base to instruct, the share of recognised entities that are unreachable at top-p 0.9 rises from 54.4% to 93.8%, and the share of those lost at the first piece from 32% to 74%. Bars: 95% cluster-bootstrap intervals.

token, while the base model starts them and loses them later.

Mass and rank. For OLMo-2, top-k RBU barely moves across stages (46.8% to 49.1% at $k = 50$) while top-p RBU rises steeply: post-training changed how much probability the head takes more than the order of the candidates. On the full TailCat, too, OLMo-2-Instruct sits close to the base models under top-k and far above them under top-p and min-p (Figure 4). Qwen2.5 does not show this pattern. Its top-k 50 RBU rises from 42.7% to 63.7%, so its post-training also reorders candidates. Truncation by mass (top-p, min-p) is the larger effect in both families, but not the only one.

6 What it does to measurement

Let $q_S(e)$ be the probability that one sample under sampler S begins with entity e. After N samples the expected number of distinct entities seen is $\sum_e [1 - (1 - q_S(e))^N]$, which approaches the size of the reachable set $\{e : q_S(e) > 0\}$ as N grows. Unseen-species estimators, from Good-Turing to Chao1 (Chao, 1984) and ACE (Chao and Lee, 1992), aim at that support size. Applied to truncated samples they can at best recover the reachable set, never the recognised set. Truncation thus sets a ceiling. Below it, a sampler may still find few names, because the distribution itself puts little mass on them. Li et al. (2025) observed that nucleus sampling lowers their knowledge estimates and sampled without it; the ceiling explains why, and top-k and the shipped chains impose one too.

Zero or merely tiny? Without truncation, a deleted name has a small but positive probability. In a post hoc analysis we compute it from the stored teacher-forced log-probabilities. This reading is well calibrated. In 11,000 untruncated pilot samples per model, the models gave 3,466 member answers against 3,653 predicted (Qwen2.5-7B-Instruct) and 4,042 against 4,069 (OLMo-2-Instruct). For most deleted names the probability is tiny. On the full TailCat, the median recognised name that top-p 0.9 deletes from Qwen2.5-7B-Instruct has probability 1.4×10^{-11}; seeing it once would take about 7×10^{10} samples. Only 1.3% of the deleted names would appear in 1,000 untruncated samples, and 12.3% in a million. The burial comes from post-training (Section 5); truncation seals it, and its cost is the ceiling. Under top-p 0.9 no amount of sampling finds more than 1,252 of the 15,514 recognised names across the 277 prompts. At 1,000 samples per prompt, truncated sampling finds between 284 and 417 of them in expectation, and untruncated sampling 575. At a million samples per prompt the truncated count stays at or below 863, while untruncated sampling reaches 2,587; it passes the ceiling after about 24,000 samples per prompt. Under the shipped sampler the ceiling is 194 names, and untruncated sampling at the same temperature passes it after about 49 samples per prompt. This last calculation omits the repetition penalty, so it is approximate.

The estimator gap, split. With 1,000 samples per category under Qwen2.5-7B-Instruct's shipped sampler on the full TailCat, the samples name 141

distinct members. The species set is our capped member list (at most 150 per category). Chao1 estimates 141 and ACE 141, while 231 members are reachable, so at this sample size Chao1 recovers 61.0% of the reachable set. The model recognises 15,514 members, and truncation seals 15,320 of them exactly. Without truncation, the same number of samples would find about 399 recognised names (approximate, as above), where truncation caps it at 194. The gap between 141 and 15,514 is therefore mostly concentration and partly truncation (Figure 3). Under top-p 0.9 the samples name 432 members, Chao1 estimates 470, 2,134 members are reachable and 15,514 recognised.

Reachability is computed for the canonical tokenisation of the listed name forms. A few sampled members fall outside the computed reachable set: 6 under the shipped sampler and 2 under top-p 0.9 for Qwen2.5-7B-Instruct, and 5 for OLMo-2-Instruct. In the cases we inspected the model wrote the name through a different tokenisation, or as a case variant of a listed form ("Universidad de los Andes" for the listed "Universidad de Los Andes"). No sample matched the token sequence of a robustly unreachable answer.

7 Who loses

Popularity. We fit a logistic model of unreachability on $\log_{10}(1+\text{sitelinks})$ for recognised entities, with generalised estimating equations clustered by category (Liang and Zeger, 1986). For Qwen2.5-7B-Instruct each tenfold increase in Wikipedia language coverage multiplies the odds of being unreachable by 0.15 (95% CI 0.12 to 0.18); for OLMo-2-Instruct by 0.10. The same model for recognition among all members gives 2.14 for Qwen2.5-7B-Instruct. Figure 5 (Appendix D) shows the same pattern by popularity bin: unreachability falls steadily with popularity, while recognition rises only in the most popular bin. Rarer names also tend to have more tokens, and so more chances to be cut. In a post hoc check that adds the label's token count and entity-type fixed effects, the odds ratio is 0.14 (0.11 to 0.18) for Qwen2.5-7B-Instruct and 0.09 for OLMo-2-Instruct, and each extra token raises the odds by 6% to 9%. Under Qwen's shipped sampler the popularity effect is stronger still (odds ratio 0.019 without controls, 0.015 with them).

Regions and types. Appendix D breaks RBU down by world region and entity type. RBU is not higher for categories from South Asia, Africa or Southeast Asia than for European or North American ones; it is lower. For Qwen2.5-7B-Instruct at top-p 0.9 it is 87.3%, 87.1% and 85.8% for South Asian, Southeast Asian and African categories, against 96.8% and 99.5% for European and North American ones. We did not predict this. After seeing it we tested one explanation, post hoc: Western categories have more recognised members competing for the same few first tokens. That effect is real but small (odds ratio 1.40 per e-fold more recognised members), and controlling for it leaves 93% of the gap on the log-odds scale. With composition, first-token width, popularity, name length, aliases and entity type all controlled, the odds of unreachability for these categories remain 0.25 times the Western odds (95% CI 0.11 to 0.55; OLMo-2-Instruct 0.15). The gap is already present in the base models (odds ratio 0.41 for Qwen2.5-7B and 0.31 for OLMo-2), these prompts leave more names reachable per category, not only a larger share, and post-training removes a smaller share of base-reachable names there (78.4% against 92.7% for Qwen2.5, 71.6% against 87.3% for OLMo-2). We do not know why. This is not a sign that non-Western knowledge is safe: the models recognise fewer of these names in the first place (31.4% of South Asian and 32.7% of African members, against 50.8% of European ones), a loss our test does not measure. We make no regional claim about who loses. The robust pattern is within categories: the rarer a recognised entity, the more likely it is unreachable.

8 A targeted fix

Li et al. (2025) call for moving from estimating knowledge to extracting it: surfacing what a model encodes but rarely generates. If most failures happen at the first token, a natural fix is to switch truncation off for the first answer token only and keep the shipped sampler afterwards. We tested this against a sweep of standard samplers on the pilot (1,000 samples per category per sampler), measuring RBU, the number of distinct real members produced and the invented-name rate (answers that match no member of the category in Wikidata and no foil).

The fix fails our pre-registered bar of recovering half of the unreachable recognised entities. It recovers 12.3% for Qwen2.5-7B-Instruct (RBU from 98.9% to 86.7%, distinct members sampled

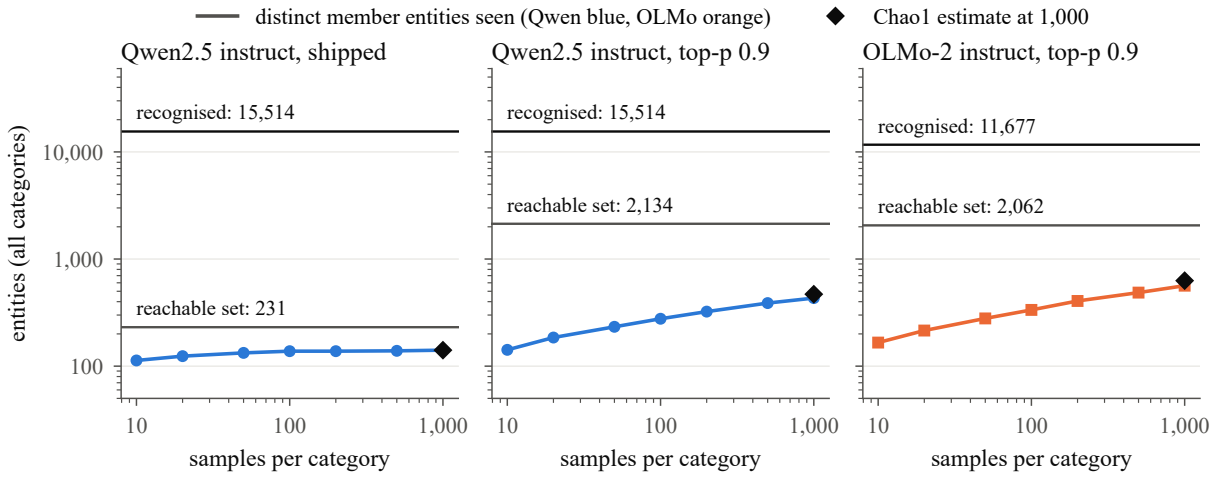


Figure 3: Sample-based knowledge estimates see the reachable set, not what the model recognises (full TailCat, template 0, 1,000 samples per category). Curves: distinct member entities seen as samples grow; diamond: the Chao1 richness estimate from all samples; lines: the reachable set and the recognised set from the exact test. With the shipped sampler, Qwen yields 141 distinct members and Chao1 estimates 141, against 231 reachable and 15,514 recognised.

from 15 to 41, invented-name rate -0.8 points) and 23.7% for OLMo-2-Instruct (invented-name rate $+3.1$ points). The reason is instructive. The first failing token is usually the first token, but most names fail again later under the same sampler (as Montpelier does in the introduction), so relaxing one position rescues only names whose every failure is at the start.

The full frontier is the more useful result (Appendix E). Removing truncation entirely makes every recognised entity reachable at a cost of 56.7% invented answers against 52.4% under the shipped sampler for Qwen2.5-7B-Instruct; top-p 0.99, top-k 100 and epsilon sampling lie in between (Figure 6). For most names this recovery is formal: they become reachable at probabilities near 1.4×10^{-11} (Section 6). The practical measure is how many distinct real members a sampler produces. In 1,000 samples per pilot category, untruncated sampling produced 75 distinct members against 15 under the shipped sampler, and first-token relaxation 41. The automatic invented-name rate overstates invention: it counts real members spelled differently, written in another script or missing from Wikidata, and wrong answers, as invented. With looser matching the reference rates fall by 11.2 points for Qwen2.5-7B-Instruct and 12.5 for OLMo-2-Instruct, while the differences between samplers barely change (relaxation: $+0.5$ and $+2.3$ points). Matching labels in every language Wikidata holds gives **[pdInventedMultilingual]** A hand audit of these answers is pending: **[pdAuditInvented]**

A second fix keeps truncation out of the answer and lets the model check its own answers with the recognition probe, keeping only answers it endorses. It was pre-registered with the same bar; because that bar counts formal recovery, we also report distinct recognised members per 1,000 samples: **[pdSelfCheckFix]** Generating and then verifying is an established pattern, and we claim no novelty for it. Its risk is that post-training also makes confident errors on long-tail questions far more common (Huang et al., 2026), and such errors may pass a model's check of its own answers.

9 Related work

Truncation and what it removes. Nucleus sampling (Holtzman et al., 2020), epsilon and eta sampling (Hewitt et al., 2022) and min-p (Nguyen et al., 2025) all give up the tail of the distribution for fluency, and top-p can harm factuality (Lee et al., 2022). The closest work is Awad et al. (2026). They use the per-token survival test we also use. They ask which mid-frequency English words remain reachable in book continuations, compare seven base and aligned pairs, and tabulate shipped defaults. Aligned models erase more words in most families, by a small average margin. Garces Arias et al. (2026) show that human-chosen tokens often fall outside common truncation sets. We change the object and the question. We test entities the model recognises against near-miss foils, in the answer to a knowledge prompt, under the exact de-

ployed processor chain, with a numeric margin and a check against real samples. We then ask why post-training changes reachability, stage by stage, and what follows for knowledge estimates. Appendix F compares the two studies point by point.

Collapse at the answer slot. Instruction-tuned models give one answer where a distribution is wanted, with a large gap between the top two logits at the answer position (Jang et al., 2026). Across 105 models, single-answer category prompts such as "Name a tree" converge on one answer, most strongly for heavily post-trained assistants (Parikh, 2026). Post-training also multiplies confident errors on long-tail factual questions (Huang et al., 2026). Our first-token result connects this concentration to an exact consequence: the recognised entities it pushes out of the kept set have probability zero.

Knowing without saying. Gekhman et al. (2025) find that a model can rank the gold answer first internally and never generate it in 1,000 samples drawn at temperature 1, without analysing truncation; Calderon et al. (2026) find that recall, not encoding, is the bottleneck for parametric factuality. Long-tail knowledge is harder to learn (Kandpal et al., 2023) and to trust (Mallen et al., 2023); a recent survey names decoding-time truncation as a possible cause without measuring it (Badhe et al., 2026). We add the zero-probability case on top of this low-probability one: for names that truncation seals, no amount of sampling helps.

Homogenisation and knowledge collapse. Post-training reduces output diversity (Kirk et al., 2024); open-ended answers converge within and across models (Jiang et al., 2025) and across languages (Davey et al., 2026); diversity collapses at identifiable post-training stages (Karouzos et al., 2026); and prompting can recover suppressed diversity (Zhang et al., 2026). Kalavasis et al. (2025) prove a trade-off between breadth and hallucination. Peterson (2025) argues that models generating towards the centre of the distribution can narrow what people come to know, Wright et al. (2025) measure the epistemic diversity of model claims, and training on generated text loses the tails (Shumailov et al., 2024) and linguistic diversity (Guo et al., 2024). These works measure diversity at a fixed decoding setting; Jiang et al. (2025) find homogeneity persists under min-p. We separate, exactly, the part of collapse that is a decoding artefact.

Estimating what a model knows from samples. Li et al. (2025) estimate unseen knowledge with smoothed Good-Turing and observe that nucleus sampling lowers their estimates, so they sample without it. We show that under truncation any such estimator meets a hard ceiling, the reachable set, and split the gap to the recognised set into that ceiling and the concentration beneath it, using Chao1 (Chao, 1984) and ACE (Chao and Lee, 1992) as examples.

Limitations

Recognition. Our recognition test is a proxy. Its yes/no part separates members from near-miss foils with AUROC between 0.79 and 0.85 across our models, below the 0.85 we set in advance; chat models answer "Yes" readily to plausible foils. The four-way choice and the per-category foil threshold limit false positives, and the results hold on the categories that pass on their own and under a second probe wording (Table 3), but RBU is defined relative to this test. **[pdAskDirectly]**

Prompts. Every main result uses prompts that ask for one name. **[pdListPrompt]**

Sampler order. "Shipped" means what `generate()` in `transformers` 5.22.2 does with each model's generation configuration. Engines such as vLLM, llama.cpp and SGLang order the processors differently, so their default chains need their own check.

Tokenisation. Reachability is computed for the canonical tokenisation, in context, of the listed name forms. A name can sometimes be produced through another token path or as an unlisted case variant. In the full-TailCat sample check of Qwen2.5-7B-Instruct, only 10 distinct entities (shipped sampler) and 34 (top-p 0.9) were ever produced solely through a non-canonical path, and in the estimator samples 6, 2 and 5 sampled members fall outside the computed reachable set (Section 6). Canonical scoring therefore overstates RBU by a negligible amount.

Numerics. Exact zero depends on how logits are computed. Our robust verdict holds for any computation within a calibrated margin of ours; implementations further away (other GPUs, quantisation) need their own calibration. The untruncated probabilities for the shipped sampler omit its repetition penalty and are approximate.

What truncation costs. For most deleted names the cost of truncation is a ceiling on discovery, not lost probability mass: without truncation they would still be very unlikely. Claims about the burial beneath the ceiling rest on teacher-forced log-probabilities, which match real samples on the pilot but were not checked against samples at full scale.

Invented answers. The invented-name rate is automatic. It counts spelling variants, real names missing from Wikidata and names in other scripts as invented, so the fix experiments depend on the hand audit and the multilingual matching.

Scope. Two model families at 7B scale, English prompts and Wikidata categories. **[pdThirdFamily]** Gemma-2 was not run, and the multilingual analysis planned for Hindi, Tamil and Bengali prompts is future work. Wikidata membership is incomplete, which affects the invented-name rate more than RBU.

Ethical considerations

Risks. Wider sampling recovers knowledge but also raises the rate of invented answers, so any fix should be paired with a check of what it produces. TailCat lists public names from Wikidata, including living people; it holds labels, aliases and sitelink counts only, and no personal data beyond what Wikidata publishes.

Licences and intended use. Qwen2.5-7B, Qwen2.5-7B-Instruct, Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct and all OLMo-2-1124-7B checkpoints are released under Apache 2.0; Qwen2.5-3B-Instruct, used only in the scale check, under the Qwen Research License. We use all models for research only and do not redistribute weights. TailCat is built from Wikidata structured data, which is released under CC0, and we release it under CC0.

Compute and annotation. All experiments used 5.28 GPU-hours on one NVIDIA RTX 5090 and a shared NVIDIA A100. Two small audits are done by the authors without payment: **[pdAuditDetails]**

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A Numeric margin

For each model we sample 500 answers per pilot category and sampler with a separate seed,

compute for each distinct answer the logit shift at which its reachability verdict flips (by bisection), and set ϵ to the smallest value on the grid $\{0.25, 0.5, 0.75, 1, 1.5, 2, 3, 4, 6\}$ that covers every sampled answer. For Qwen2.5-7B-Instruct, 1,685 distinct sampled answers gave 63 exact-verdict contradictions and none at $\epsilon = 0.75$; the 99.9th percentile of the relevant logit gap was 0.72. In the Qwen sample check, 1 sampled answer would have contradicted the exact verdict and none contradicted the robust one. Batch jitter is visible even within one pass: Montpelier and Montgomery share their first token after the same prompt, yet the probability mass ahead of that token is 0.909 in one row of the batched pass and 0.911 in another.

B Recognition validity

category	model	AUROC	4-way acc.	recognised	foils passing
assamese_writers	instruct	0.716	0.711	18	5.0
bengali_film_directors	instruct	0.810	0.888	177	5.0
brazil_universities	instruct	0.979	0.835	177	5.1
finland_lakes	instruct	0.881	0.818	280	5.0
japanese_novelists	instruct	0.987	0.978	384	5.0
kerala_rivers	instruct	0.806	0.500	35	5.6
nepal_mountains	instruct	0.776	0.550	50	5.5
nigerian_painters	instruct	0.891	0.644	31	6.7
tamil_poets	instruct	0.802	0.740	58	5.4
tamilnadu_hindu_temples	instruct	0.738	0.555	73	5.0
us_state_capitals	instruct	0.874	0.939	27	6.8
ALL (pooled)	instruct	0.835	0.767	1310	–
assamese_writers	base	0.677	0.651	21	5.0
bengali_film_directors	base	0.833	0.819	153	5.0
brazil_universities	base	0.978	0.839	179	5.1
finland_lakes	base	0.799	0.845	244	5.0
japanese_novelists	base	0.987	0.873	346	5.0
kerala_rivers	base	0.695	0.510	23	5.6
nepal_mountains	base	0.746	0.544	51	5.5
nigerian_painters	base	0.824	0.630	22	6.7
tamil_poets	base	0.856	0.699	55	5.4
tamilnadu_hindu_temples	base	0.758	0.482	83	5.0
us_state_capitals	base	0.948	0.939	38	6.8
ALL (pooled)	base	0.814	0.728	1215	–

Table 2: Recognition validity per pilot category (Qwen2.5-7B pair, main probe). Foils passing: share of foils at or above the category threshold.

C Walk-through examples

Each table shows one recognised entity token by token under top-p 0.9 (template 0): the chat model's verdict first, then the base model's.

Shintarō Ishihara (japanese_novelists, Qwen2.5-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	`'Sh'`	0.998 / cut
	`'int'`	0.952 / cut
	`'ar'`	0.970 / cut
	`'ō'`	0.000 / kept
	`' Ish'`	0.000 / kept
	`'ih'`	0.000 / kept
	`'ara'`	0.000 / kept
base	`' Sh'`	0.799 / kept
	`'int'`	0.819 / kept
	`'ar'`	0.971 / cut
	`'ō'`	0.000 / kept
	`' Ish'`	0.000 / kept
	`'ih'`	0.000 / kept
	`'ara'`	0.000 / kept

Himalchuli (nepal_mountains, Qwen2.5-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	`'H'`	0.999 / cut
	`'imal'`	0.000 / kept
	`'ch'`	0.000 / kept
	`'uli'`	0.000 / kept
base	`' Himal'`	0.847 / kept
	`'ch'`	0.936 / cut
	`'uli'`	0.000 / kept

Federal University of Science and Innovation of Minas Gerais (brazil_universities, OLMo-2-7B)

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	`'Federal'`	0.914 / cut
	`' University'`	0.000 / kept
	`' of'`	0.000 / kept
	`' Science'`	1.000 / cut
	`' and'`	0.000 / kept
	`' Innovation'`	1.000 / cut
	`' of'`	0.998 / cut
	`' Min'`	0.887 / kept
	`'as'`	0.000 / kept
	`' G'`	0.000 / kept
	`'erais'`	0.000 / kept
base	`' Federal'`	0.261 / kept
	`' University'`	0.000 / kept
	`' of'`	0.000 / kept
	`' Science'`	0.995 / cut
	`' and'`	0.000 / kept
	`' Innovation'`	1.000 / cut
	`' of'`	0.922 / cut
	`' Min'`	0.410 / kept
	`'as'`	0.000 / kept
	`' G'`	0.000 / kept
	`'erais'`	0.000 / kept

State University of Feira de Santana (brazil_universities, OLMo-2-7B)

run	model	sampler	categories	recognised	RBU-mean	RBU-all
pilot (all)	Qwen2.5-7B-Instruct	p0.90_T1	11	1,310	95.1	94.0
pilot (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	p0.90_T1	5	899	98.2	97.9
pilot (all)	Qwen2.5-7B-Instruct	shipped_hf	11	1,310	98.9	98.6
pilot (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	shipped_hf	5	899	99.5	99.2
pilot (all)	Qwen2.5-7B	p0.90_T1	11	1,215	52.4	47.0
pilot (AUROC $\geq$ 0.85)	Qwen2.5-7B	p0.90_T1	5	829	54.5	48.5
pilot, second probe (all)	Qwen2.5-7B-Instruct	p0.90_T1	11	1,234	94.8	93.8
pilot, second probe (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	p0.90_T1	3	588	98.8	98.6
pilot, second probe (all)	Qwen2.5-7B-Instruct	shipped_hf	11	1,234	98.6	98.2
pilot, second probe (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	shipped_hf	3	588	99.2	98.8
pilot, second probe (all)	Qwen2.5-7B	p0.90_T1	11	1,217	52.5	47.1
pilot, second probe (AUROC $\geq$ 0.85)	Qwen2.5-7B	p0.90_T1	3	562	55.5	50.0
pilot (all)	OLMo-2-1124-7B-Instruct	p0.90_T1	11	1,124	93.8	91.9
pilot (AUROC $\geq$ 0.85)	OLMo-2-1124-7B-Instruct	p0.90_T1	4	837	98.6	97.7
pilot (all)	OLMo-2-1124-7B	p0.90_T1	11	1,032	54.4	47.0
pilot (AUROC $\geq$ 0.85)	OLMo-2-1124-7B	p0.90_T1	4	762	55.5	47.8
full TailCat (all)	Qwen2.5-7B-Instruct	p0.90_T1	277	15,514	91.9	89.7
full TailCat (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	p0.90_T1	139	11,023	93.1	91.2
full TailCat (all)	Qwen2.5-7B-Instruct	shipped_hf	277	15,514	98.8	98.5
full TailCat (AUROC $\geq$ 0.85)	Qwen2.5-7B-Instruct	shipped_hf	139	11,023	99.1	98.8
full TailCat (all)	Qwen2.5-7B	p0.90_T1	277	16,302	55.0	50.1
full TailCat (AUROC $\geq$ 0.85)	Qwen2.5-7B	p0.90_T1	139	11,264	58.2	53.2
full TailCat (all)	OLMo-2-1124-7B-Instruct	p0.90_T1	277	11,677	91.2	88.5
full TailCat (AUROC $\geq$ 0.85)	OLMo-2-1124-7B-Instruct	p0.90_T1	104	7,030	93.2	90.9
full TailCat (all)	OLMo-2-1124-7B	p0.90_T1	277	12,324	62.7	56.3
full TailCat (AUROC $\geq$ 0.85)	OLMo-2-1124-7B	p0.90_T1	104	7,328	66.2	60.2

Table 3: Robust RBU on all categories and on the categories where the instruct model's yes/no probe reaches AUROC 0.85 on its own (the same categories for both models of a pair), and under a second probe wording with worked examples. RBU-mean averages over prompt wordings; RBU-all counts an entity only if it is unreachable under every wording.

model	token	mass ahead (G, $T = 1$) / kept at $p = 0.9$
chat	`'State'`	0.993 / cut
	`' University'`	0.000 / kept
	`' of'`	0.000 / kept
	`' Fe'`	1.000 / cut
	`'ira'`	0.000 / kept
	`' de'`	0.000 / kept
	`' Santana'`	0.000 / kept
base	`' State'`	0.777 / kept
	`' University'`	0.000 / kept
	`' of'`	0.000 / kept
	`' Fe'`	0.915 / cut
	`'ira'`	0.000 / kept
	`' de'`	0.000 / kept
	`' Santana'`	0.000 / kept

D Breakdowns

region	OLMo-2-1124-7B	OLMo-2-1124-7B-Inst	Qwen2.5-7B	Qwen2.5-7B-Inst
africa	49.7	82.5	42.5	85.8
east_asia	70.1	97.5	65.2	98.9
europe	76.6	95.8	69.1	96.8
latin_america	60.4	91.5	56.6	91.5
north_america	79.7	99.0	65.6	99.5
south_asia	49.7	83.9	43.0	87.3
southeast_asia	53.0	86.1	42.7	87.1
west_asia	61.4	91.4	53.6	90.1

Table 4: Robust RBU (%) at top-p 0.9 by region, full TailCat.

type	OLMo-2-1124-7B	OLMo-2-1124-7B-Inst	Qwen2.5-7B	Qwen2.5-7B-Inst
architects	64.8	88.6	53.2	87.1
composers	54.6	87.6	48.5	86.4
film_directors	57.8	91.6	50.8	88.4
films	44.9	84.8	53.6	93.1
football_clubs	65.6	93.2	71.4	96.4
islands	72.4	91.4	58.4	89.3
lakes	73.3	96.1	64.2	94.4
mountains	77.5	95.8	63.1	95.1
museums	69.7	97.2	66.0	98.0
novelists	59.6	88.4	54.5	92.4
painters	54.4	85.4	42.6	82.5
poets	57.8	90.4	51.0	87.4
rivers	69.4	91.2	49.7	96.8
universities	62.4	96.8	50.2	99.1
writers_by_language	39.7	79.7	38.9	82.5

Table 5: Robust RBU (%) at top-p 0.9 by type, full TailCat.

E Mitigation frontier

F Comparison with the closest work

G Deviations from the pre-registered plan

We fixed thresholds and analyses before running the experiments and logged every later change with its reason. The changes that affect this paper:

- **Recognition bar.** The plan required pooled yes/no AUROC of at least 0.85 on the pilot. The pilot reached 0.835, and a second probe wording did not raise it. We proceeded by a recorded decision, kept the first probe as the main analysis, and report the second probe and the categories that pass on their own as checks (Section 3, Table 3).

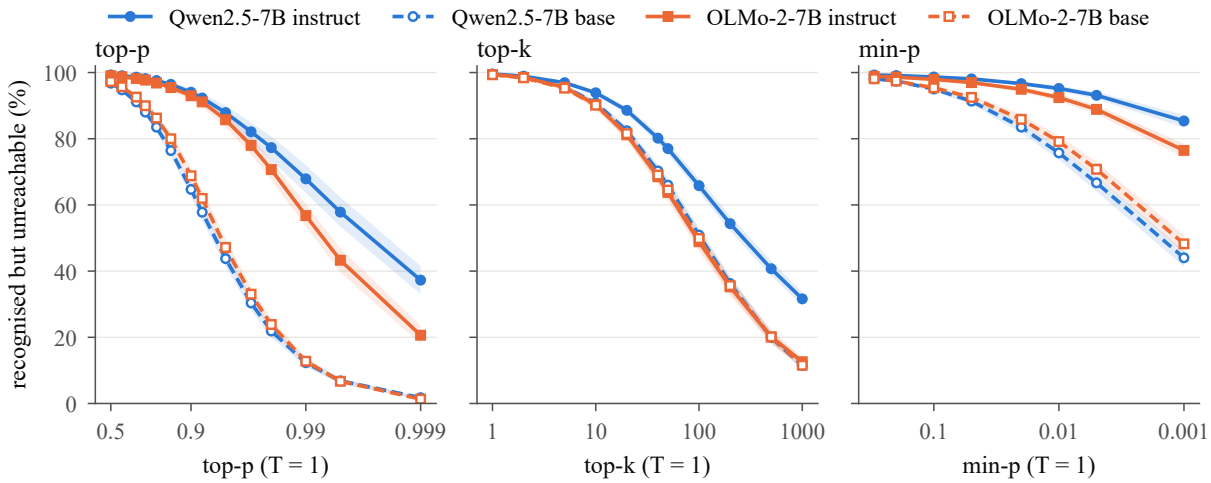


Figure 4: Share of recognised entities that are unreachable (RBU, exact verdict) as each truncation is loosened, full TailCat (277 categories), mean over three wordings with 95% cluster-bootstrap bands. Left to right within each panel the truncation gets weaker. Solid, filled: post-trained (instruct) models; dashed, open: their base models. At top-p 0.9 the Qwen instruct model leaves 94.0% unreachable against 64.7% for its base model.

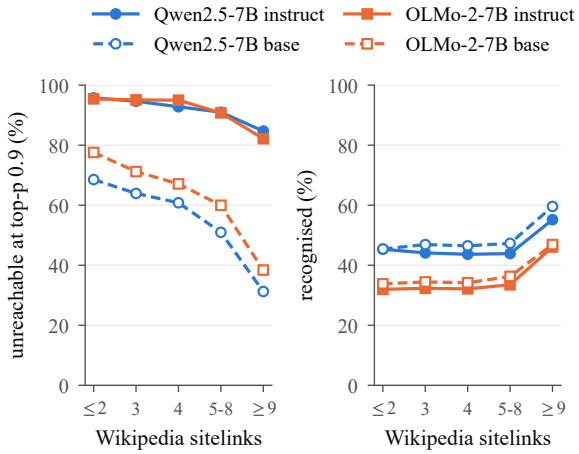


Figure 5: Who loses, on the full TailCat (33,437 members). Entities are binned by Wikipedia sitelinks (popularity quintiles; ties make the bins unequal). Left: share of recognised entities that are unreachable at top-p 0.9 (robust verdict). Right: share recognised. For the Qwen instruct model, unreachability falls from 95.8% in the least linked bin to 84.7% in the most linked bin.

- **Head control.** The US-state-capitals control was judged on the base model, which reaches every capital; the instruct model's result is reported as a finding (Section 4).

- **Headline statistic.** The plan named RBU-all, the share unreachable under every wording, as the conservative headline. The text reports RBU-mean, which uses every wording; Table 1 gives RBU-all beside it.

- **Regression.** The plan named a mixed-effects logistic model with a random intercept per category. We fitted generalised estimating equations clustered by category instead, because our statistics library has no maximum-likelihood mixed logistic model; both account for clustering by category.

- **Multiple comparisons.** Holm correction over the 6 pre-registered tests that carry p-values (the two base-against-final sign tests and four popularity slopes) changes no decision: the largest adjusted p-value is $< 10^{-74}$.

- **Post hoc analyses.** Four analyses were chosen after seeing results and are labelled post hoc in the text: how likely deleted names would be without truncation (Section 6), the mechanism correlation within models and categories (Section 5), the popularity effect with length and type controls, and the region analysis (Section 7). No gate depends on them.

- **Invented-name metric.** An internal review found sampled answers in Chinese, Japanese and Korean script, some of them real members. We therefore match labels in every language Wikidata holds. We made the change before running the second fix experiment and report the first fix under both metrics (Section 8).

H Reproducibility

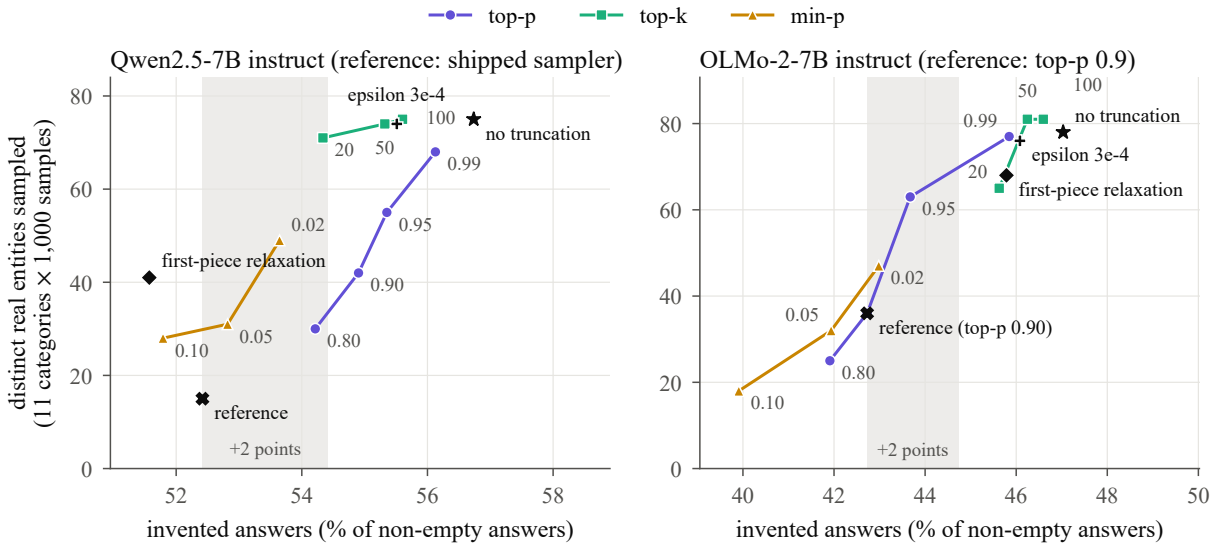


Figure 6: The mitigation frontier (pilot, 11 categories, 1,000 samples per category and sampler). Each point is a sampler: distinct real member entities it produced against the share of answers that match no real member ("invented"; includes spelling variants and real names missing from Wikidata, see the hand audit). Lines join settings of one sampler family, labelled by their parameter. Shaded: within 2 points of the reference, the band gate G3 allowed. For Qwen, removing truncation raises distinct entities from 15 to 75 at 4.3 more points of invented answers.

model	sampler	RBU	recovered	distinct members	invented	Δ invented (pts)
Qwen2.5-7B-Instruct	untruncated_T1	0.0	100.0	75	56.7	+4.3
Qwen2.5-7B-Instruct	p0.80_T1	97.7	1.2	30	54.2	+1.8
Qwen2.5-7B-Instruct	p0.90_T1	95.3	3.6	42	54.9	+2.5
Qwen2.5-7B-Instruct	p0.95_T1	91.1	7.9	55	55.4	+2.9
Qwen2.5-7B-Instruct	p0.99_T1	74.0	25.2	68	56.1	+3.7
Qwen2.5-7B-Instruct	k20_T1	79.6	19.5	71	54.3	+1.9
Qwen2.5-7B-Instruct	k50_T1	62.2	37.1	74	55.3	+2.9
Qwen2.5-7B-Instruct	k100_T1	46.6	52.8	75	55.6	+3.2
Qwen2.5-7B-Instruct	minp0.02_T1	95.0	3.9	49	53.6	+1.2
Qwen2.5-7B-Instruct	minp0.05_T1	97.2	1.7	31	52.8	+0.4
Qwen2.5-7B-Instruct	minp0.10_T1	98.2	0.6	28	51.8	-0.6
Qwen2.5-7B-Instruct	eps3e-4_T1	75.9	23.2	74	55.5	+3.1
Qwen2.5-7B-Instruct	reference	98.9	0.0	15	52.4	+0.0
Qwen2.5-7B-Instruct	relax_first	86.7	12.3	41	51.6	-0.8
OLMo-2-1124-7B-Instruct	untruncated_T1	0.0	100.0	78	47.0	+4.3
OLMo-2-1124-7B-Instruct	p0.80_T1	97.2	-3.3	25	41.9	-0.8
OLMo-2-1124-7B-Instruct	p0.90_T1	94.0	0.0	36	42.7	+0.0
OLMo-2-1124-7B-Instruct	p0.95_T1	89.8	4.5	63	43.7	+0.9
OLMo-2-1124-7B-Instruct	p0.99_T1	58.6	37.7	77	45.8	+3.1
OLMo-2-1124-7B-Instruct	k20_T1	70.5	25.1	65	45.6	+2.9
OLMo-2-1124-7B-Instruct	k50_T1	49.6	47.3	81	46.2	+3.5
OLMo-2-1124-7B-Instruct	k100_T1	35.1	62.6	81	46.6	+3.9
OLMo-2-1124-7B-Instruct	minp0.02_T1	93.9	0.2	47	43.0	+0.3
OLMo-2-1124-7B-Instruct	minp0.05_T1	96.8	-2.9	32	41.9	-0.8
OLMo-2-1124-7B-Instruct	minp0.10_T1	98.3	-4.5	18	39.9	-2.8
OLMo-2-1124-7B-Instruct	eps3e-4_T1	64.4	31.5	76	46.1	+3.4
OLMo-2-1124-7B-Instruct	reference	94.0	0.0	36	42.7	+0.0
OLMo-2-1124-7B-Instruct	relax_first	71.7	23.7	68	45.8	+3.1

Table 6: Mitigation frontier on the pilot (template 0, 1,000 samples per category per sampler). Reference: shipped sampler (Qwen2.5), top-p 0.9 (OLMo-2, which ships none). Invented: automatic English-label matching (Section 8).

	Awad et al. (2026)	this paper
Object	mid-frequency English words in book continuations	named entities in the answer to a knowledge prompt
What counts as known	not tested (the score measures lexical survival, not knowledge)	yes/no against near-miss foils plus a four-way choice
Test	per-token survival, one teacher-forced pass	the same test, run through each model's deployed processor chain, with aliases and critical values
Shipped defaults	tabulated; nearest evaluated temperature used where the documented one was not evaluated	the exact generation configuration, including repetition penalty
Post-training	base against final aligned models	base and three OLMo-2 post-training stages
Mechanism	not measured	the cut moves to the first token; first-token concentration tracks RBU
Measurement consequence	none	unseen-species estimators recover at most the reachable set; exact truncation gap
Who loses	one frequency band	popularity within 277 categories from 24 countries
Numerics and validation	not discussed	calibrated margin; checked against real samples
Fix	discussed	tested: first-token relaxation and a frontier of samplers

Table 7: What Awad et al. (2026) do and what this paper adds. Their test, their base-against-aligned comparison and their link to recursive homogenisation come first; our contribution is the knowledge setting, the stage-wise mechanism, the measurement consequence and the scale of the who-loses analysis.

run	model	snapshot	GPU	ϵ	minutes
G0_5090	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	2.3
G0_5090	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	4.6
G1	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	2.4
G1	OLMo-2-1124-7B-DPO	`e34ea60`	GeForce RTX 5090	0.5	2.4
G1	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	2.4
G1	OLMo-2-1124-7B-SFT	`1de02c0`	GeForce RTX 5090	0.5	2.4
P3_robust	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	3.3
P3_robust	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	2.4
P3_robust	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	3.5
P3_robust	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	4.4
P3	OLMo-2-1124-7B	`7df9a82`	GeForce RTX 5090	0.5	27.6
P3	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	39.4
P3	Qwen2.5-7B	`d149729`	GeForce RTX 5090	0.75	30.0
P3	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	79.0
P3_scale	Qwen2.5-0.5B-Instruct	`7ae5576`	GeForce RTX 5090	3.0	1.7
P3_scale	Qwen2.5-1.5B-Instruct	`989aa79`	GeForce RTX 5090	0.75	1.8
P3_scale	Qwen2.5-3B-Instruct	`aa8e725`	GeForce RTX 5090	2.0	2.1
P5	OLMo-2-1124-7B-Instruct	`470b1fb`	GeForce RTX 5090	0.5	11.3
P5	Qwen2.5-7B-Instruct	`a09a354`	GeForce RTX 5090	0.75	13.6

Table 8: Runs behind every number: model snapshot (Hugging Face commit), GPU, numeric margin and wall time. Data, outputs, metric tables and the decision log are in the supplementary material; total GPU time for the project 5.28 hours.